BA ZC581 :
Applied Data Science
Module: 02

BITS Pilani

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J A Gokhale.

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Disclaimer

- The slides presented here are obtained / extracted from the authors of the prescribed books and from various other subject contributors.
- I hereby thank and acknowledge all the contributors for their material and inputs.
- I have added and modified a few slides to suit the requirements of the course.

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Module 2: Types of Analysis

Module 2 Topics

Topic 1: Descriptive

Topic 2: Predictive

Topic 3: Diagnostic

Topic 4: Prescriptive

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Types of Analytics

Types of Analysis / Analytics

Descriptive

Predictive

Diagnostic

Prescriptive

Inferential

Exploratory

Cognitive

We will study these in detail

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Topic 1: Descriptive Analysis

Descriptive Analytics

- What happened or what is happening?
- Use current data and historical data to identify the present state of business.
- Use statistical techniques to understand the data
  - Ex: Examining this Quarter's sales, we can estimate the % increase of sales over the previous quarter. A performance indicator.
- Preferred Representations:
  - Dashboards
  - Stories & Summaries

Descriptive Analytics

Descriptive Analysis

- **Purpose**: Summarizes and describes the main features of a dataset.
- **Examples**: Mean, median, mode, standard deviation, and data visualizations like bar charts and histograms.
- **Use Case**: Understanding the overall distribution and basic characteristics of the data.

Example

Consider Retail Sales Data as below:

Month	Total Sales	Transactions	Avg Basket Size
January	1,20,000	400	300
February	1,50,000	500	300
March	1,80,000	600	300
April	2,10,000	700	300

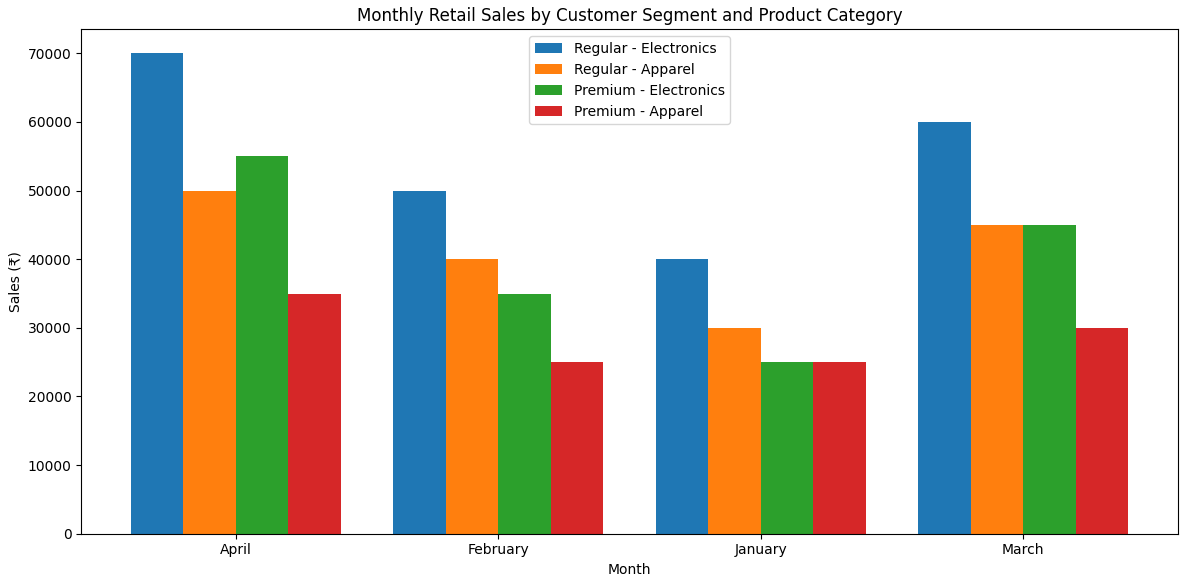
Descriptive Insights:

Total sales increased by 75% from January to April

Transaction volume grew steadily by 100 Tx per month

Basket size in April remained steady over the previous quarter, indicating stable customer spending behaviour

Visualization for Descriptive Analysis



The grouped bar chart above illustrates how sales evolved from January to April across four combinations:

Regular – Electronics

Regular – Apparel

Premium – Electronics

Premium – Apparel

Inferences / Insights from Descriptive Analysis

- Regular customers consistently drove higher sales, especially in Electronics, which grew from ₹40,000 in January to ₹70,000 in April.
- Premium Electronics sales also increased, reaching ₹55,000 in April, showing strong engagement from high-value customers.
- Apparel sales remained steady, with Regular customers contributing more than Premium ones across all months.
- The overall upward trend across all segments and categories reflects healthy growth and balanced product performance

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Topic 2: Predictive Analysis

Predictive Analytics

What will happen, or likely to happen next?

- Use current data and historical data to identify the present state to predict the future state of the business.
- Use forecasting and ML techniques to predict the future state
  - Ex: Examining this Quarter's sales, and the previous 4 quarter sales, we can estimate the sales for the next quarter. An estimator indicator.

Forecasting models like Regression

ML Models

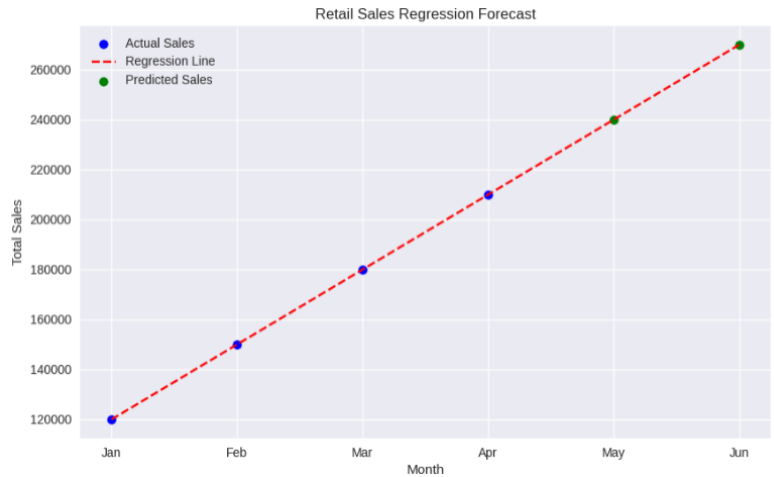
Predictive Analytics

Predictive Analysis

- **Purpose**: Uses historical data to make predictions about future events.
- **Examples**: Regression analysis, time series forecasting, and machine learning models.
- **Use Case**: Forecasting sales, predicting customer behavior, and risk assessment.

Example:

Using regression model:



- Sales for May and June are projected to be ₹240,000 and ₹270,000 respectively, continuing the steady monthly growth trend. The regression visualization confirms a strong linear pattern.

Detailed Data (Restructured)

Month	Premium Electronics	Premium Apparel	Regular Electronics	Regular Apparel	Total Sales	No of Tx	Average Bag Size
Jan	40,000	20,000	30,000	30,000	120,000	400	300
Feb	50,000	25,000	35,000	40,000	150,000	500	300
March	60,000	30,000	45,000	45,000	180,000	600	300
April	70,000	35,000	55,000	50,000	210,000	700	300

Regression based Segment Forecast (May-June)

Segment	May Forecast	June Forecast	Monthly Growth
Premium Electronics	₹80,000	₹90,000	₹10,000
Premium Apparel	₹40,000	₹45,000	₹5,000
Regular Electronics	₹65,000	₹75,000	₹10,000
Regular Apparel	₹55,000	₹60,000	₹5,000
Total Sales	₹240,000	₹270,000	₹30,000

Predictive Insights

Premium Electronics
- Highest growth rate and largest contributor to total sales.
- Indicates strong demand and brand loyalty.
- Ideal for bundling, upselling, and exclusive launches.

Premium Apparel
- Moderate growth, possibly seasonal or trend-driven.
- Opportunity for influencer campaigns or limited-edition drops.

Regular Electronics
- Steady growth, driven by volume and affordability.
- Good candidate for promotions and EMI schemes.

Regular Apparel
- Consistent but slower growth, suggesting saturation.
- Could benefit from cross-category promotions or loyalty rewards.

Strategic Recommendations

Inventory Planning: Scale up Premium Electronics stock and logistics for projected ₹90,000 sales in June.

Marketing Focus: Prioritize Premium segments for targeted campaigns; reinforce Regular segments with value-based messaging.

Customer Engagement: Maintain basket size stability while increasing transaction volume through personalized offers

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Topic 3: Diagnostic Analysis

Diagnostic Analytics

Examine data to understand the cause behind the past outcomes to answer the question “Why did it happen?”

- Root Cause Focused
  - Goes beyond “what happened” to uncover why it happened
- Data-Driven Investigation
  - Uses correlations, comparisons, and drill-downs to identify patterns and anomalies
  - Historical data is measured against standard values and templates for comparison
- Often Follows Descriptive Analysis
  - Builds on summary statistics to explore deeper insights

Example: Sales dropped because of a 25% decline in repeat customers due to price change.

Common Techniques:

- Trend and variance analysis
- Segmentation (e.g., by region, product, customer type)
- Correlation and regression
- Root cause analysis (e.g., 5 Whys, Fishbone diagrams)

Diagnostic Analytics

Diagnostic Analysis

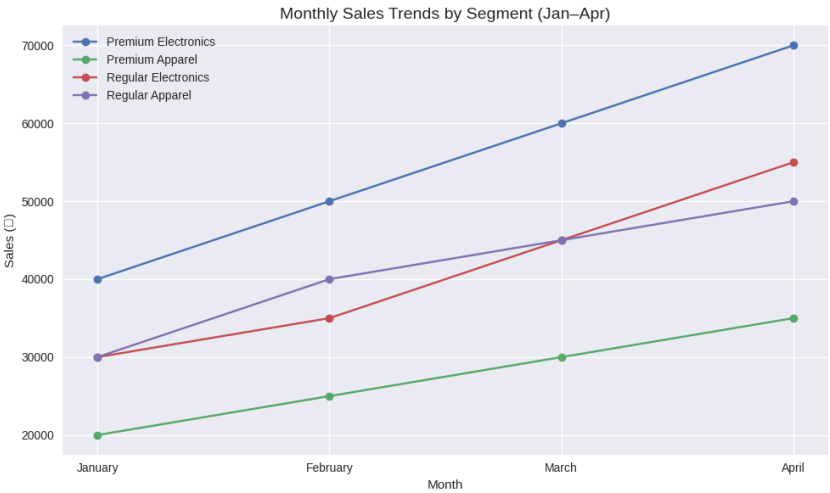
- **Purpose**: Investigates the reasons behind past outcomes or events.
- **Examples**: Root cause analysis, drill-down analysis, and correlation analysis.
- **Use Case**: Identifying the causes of specific trends or anomalies in the data.

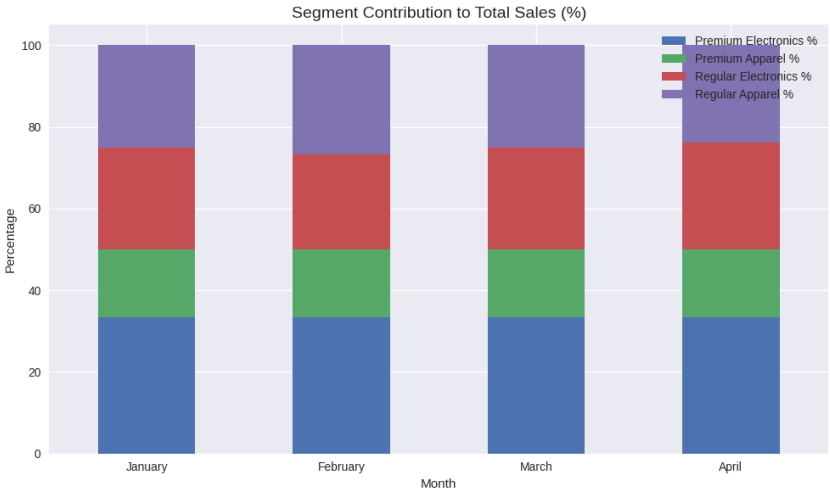
Diagnostic Analysis: Segment wise analysis

Segment Breakdown:

Segment	Jan	Feb	Mar	Apr	Growth (Jan–Apr)	Monthly Δ
Premium Electronics	₹40K	₹50K	₹60K	₹70K	+75%	₹10K
Premium Apparel	₹20K	₹25K	₹30K	₹35K	+75%	₹5K
Regular Electronics	₹30K	₹35K	₹45K	₹55K	+83%	₹8.3K
Regular Apparel	₹30K	₹40K	₹45K	₹50K	+66%	₹6.6K

Visualizations





Segment-wise performance dashboards

Segment	Jan	Apr	Growth (%)	Monthly Δ (₹)	Contribution (Apr)
Premium Electronics	₹40K	₹70K	+75%	₹10K	33.3%
Premium Apparel	₹20K	₹35K	+75%	₹5K	16.7%
Regular Electronics	₹30K	₹55K	+83%	₹8.3K	26.2%
Regular Apparel	₹30K	₹50K	+66%	₹6.6K	23.8%

Diagnostics Insights

1. Premium Electronics: Consistent High Performer

- - Strong linear growth of ₹10K/month.
- - Likely driven by high-value items and loyal customer base.
- - No anomalies or dips—ideal for scaling.

2. Premium Apparel: Under-leveraged Potential

- - Growth is steady but modest.
- - May be constrained by limited assortment or seasonal demand.
- - Diagnostic flag: low absolute contribution despite premium positioning.

3. Regular Electronics: Accelerating Momentum

- - Growth rate increased from ₹5K (Jan–Feb) to ₹10K (Mar–Apr).
- - Indicates rising interest or successful promotions.
- - Diagnostic cue: potential crossover into premium behavior.

4. Regular Apparel: Plateau Risk

- - Growth slowed from ₹10K (Feb) to ₹5K (Apr).
- - May signal saturation or lack of differentiation.
- - Diagnostic concern: stagnation despite volume potential.

Diagnostic Analysis: Strategic Suggestions

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For Premium Electronics

- Double down: Expand SKUs, offer exclusive bundles, and prioritize logistics.
- Introduce loyalty tiers or early-access programs.

For Premium Apparel

- Revitalize with seasonal drops, influencer tie-ins, or limited editions.
- Consider cross-segment bundling with electronics.

For Regular Electronics

- Promote aggressively: EMI schemes, referral bonuses, or flash sales.
- Monitor for premium migration—customers upgrading behavior.

For Regular Apparel

- Reposition or refresh: Introduce new styles, optimize pricing, or bundle with fast-moving categories.
- Use customer segmentation to identify dormant buyers.

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Topic 4: Prescriptive Analysis

Prescriptive Analytics

Recommend specific actions to achieve the objective. Answer the question "What should we do?"

- Uses Optimization, Simulation and AI to assess various scenarios.
- Scenario simulation and perform what-if analysis
- Evaluates various options opportunities and optimizes the best
- Target: To achieve 25% increase in sales during next quarter
- Evaluate options like
  - Increase advertisement budget by 10%
  - Reduce price by 5%
  - Offer selective discount of 10%
  - Bundle several products
  - Reduce Transportation cost by 20%

Prescriptive Analytics

Prescriptive Analysis

- **Purpose**: Provides recommendations for decision-making based on data analysis.
- **Examples**: Optimization models, simulation, and decision trees.
- **Use Case**: Determining the best course of action, such as inventory management and supply chain optimization.

Segment Performance Recap

Segment wise Performance

Segment	Jan	Apr	Growth (%)	Monthly Δ (₹)	Contribution (Apr)
Premium Electronics	₹40K	₹70K	+75%	₹10K	33.3%
Premium Apparel	₹20K	₹35K	+75%	₹5K	16.7%
Regular Electronics	₹30K	₹55K	+83%	₹8.3K	26.2%
Regular Apparel	₹30K	₹50K	+66%	₹6.6K	23.8%

Prescriptive Recommendations by Segment

1. Premium Electronics

- Action: Scale inventory, introduce exclusive bundles, and offer early-access programs.
- Why: Highest growth and contribution; consistent demand signals readiness for expansion.
- How: Launch premium loyalty tiers and tech upgrade campaigns.

2. Premium Apparel

- Action: Refresh product lines with seasonal drops and influencer tie-ins.
- Why: Moderate growth but low contribution; needs differentiation.
- How: Use limited editions and cross-category bundles with electronics.

3. Regular Electronics

- Action: Promote aggressively with EMI schemes and referral bonuses.
- Why: Accelerating growth; potential to rival premium performance.
- How: Highlight affordability and reliability in marketing.

4. Regular Apparel

- Action: Reposition or bundle with fast-moving categories.
- Why: Slowing growth; risk of stagnation.
- How: Introduce value packs, loyalty discounts, and style refreshes

Prescriptive Recommendations: Cross Segment Strategies

Bundling: Pair Premium Electronics with Apparel to boost weaker segments.

Loyalty Programs: Tiered rewards based on segment engagement.

Promotional Timing: Align campaigns with seasonal spikes and product launches.

Customer Segmentation: Use transaction data to target dormant or high-potential

In Essence,

To optimize future growth, prioritize Premium Electronics and Regular Electronics with targeted promotions and inventory expansion.

Prescriptive analysis recommends bundling strategies, loyalty programs, and differentiated marketing to unlock underperforming segments.

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Topic 5: Other types

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Inferential analytics

Inferential Analysis

It uses sample data to make generalizations or predictions about a larger population. It answers the question: “What can we conclude beyond this data?

- Sample-Based Reasoning
  - Draws conclusions from a subset of data, not the entire population
- Probability-Driven
  - Uses statistical techniques to estimate confidence and significance
- Supports Decision-Making
  - Helps test hypotheses, validate assumptions, and guide strategy

Example:

Observation: “In our sample of 300 customers, 60% prefer South Indian cuisine”

Inference: “We estimate that around 60% of all customers prefer South Indian cuisine, with a ±5% margin of error”

Techniques Used:

- Hypothesis testing (e.g., t-tests, chi-square)
- Confidence intervals
- Regression analysis
- ANOVA (Analysis of Variance)
- Sampling methods (random, stratified, cluster)

Inferential Observations

1. Linear Growth Across Segments

- All segments show consistent month-over-month increases.
- Suggests a systematic growth pattern, not random fluctuation.
- Implies strong operational consistency and effective customer engagement.

2. Premium vs Regular Comparison

- Premium Electronics outperforms all other segments in absolute growth and contribution.
- Regular Electronics shows comparable growth rate, indicating potential convergence in customer behavior.
- Apparel segments grow slower, suggesting category-specific constraints (e.g., seasonality, fashion cycles).

3. Electronics vs Apparel

- Electronics segments (Premium + Regular) contribute ~60% of April sales.
- Apparel segments lag, implying lower average ticket size or purchase frequency.
- Indicates a product category effect—Electronics likely benefit from higher perceived value and urgency.

4. Customer Segment Behavior

- Premium customers show higher absolute spend, but Regular customers are catching up in Electronics.
- Suggests price elasticity and value-driven adoption in Regular segments

Insights for Strategic Purposes

For Premium Electronics

- Double down: Expand SKUs, offer exclusive bundles, and reinforce loyalty programs.
- Why: High growth and consistent demand imply strong brand affinity and repeat behavior.

For Regular Electronics

- Promote aggressively: EMI schemes, referral bonuses, and value messaging.
- Why: High growth rate and convergence with premium behavior suggest untapped potential.

For Apparel Segments

- Revitalize: Introduce seasonal collections, influencer tie-ins, and cross-category bundles.
- Why: Slower growth may be due to fashion fatigue or lack of differentiation.

For Cross-Segment Strategy

- Bundle Premium Electronics with Apparel to lift weaker segments.
- Use customer segmentation to identify high-potential Regular buyers for premium upsell

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Exploratory Analysis (EDA)

Exploratory Analysis

Also called EDA, it examines datasets to uncover patterns, identify trends, discover relationships, spot anomalies, test assumptions, and generate hypotheses.

A step carried before formal modeling begins

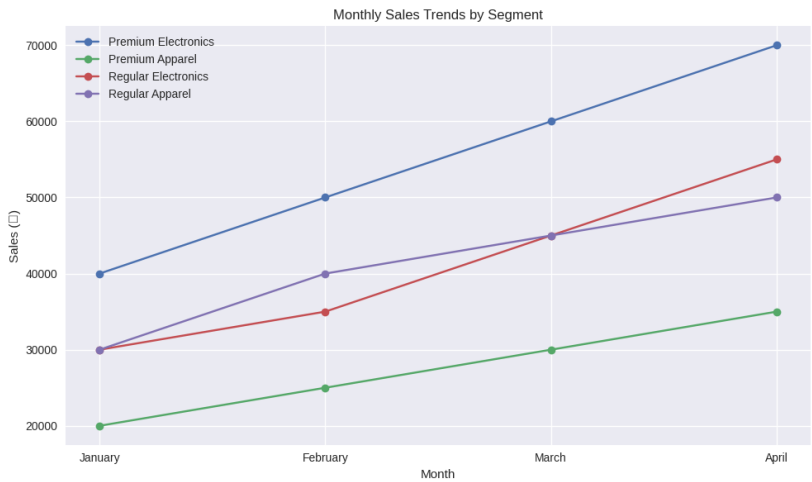
- Understand data structure, distributions, and relationships
- Detect missing values, outliers, and inconsistencies
- Identify potential features for modeling

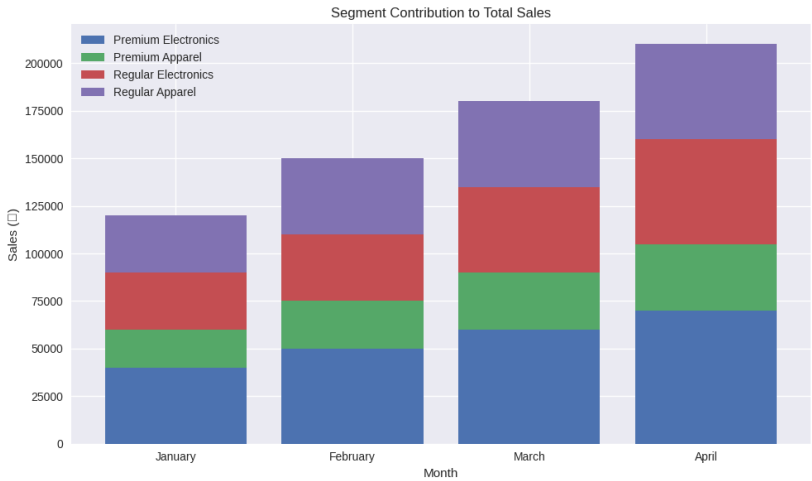
Example: Let's explore customer purchase data to see if age or location influences buying behavior.

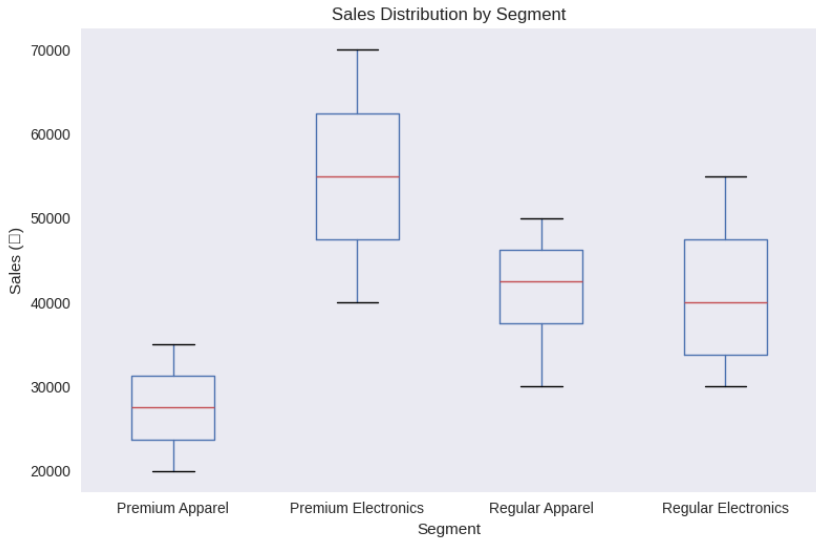
Common Techniques

- Summary statistics (mean, median, mode, std. dev.)
- Data visualizations (histograms, box plots, scatter plots, heatmaps)
- Correlation matrices
- Group-wise comparisons and aggregations
- Dimensionality reduction (e.g., PCA for high-dimensional data)

EDA in visuals







EDA Visualizations reveal clear trends and distribution patterns that support strategic planning

Key EDA Insights

- Growth Consistency: All segments grow steadily, ideal for regression-based forecasting.
- Segment Imbalance: Premium Electronics outpaces others, while Premium Apparel lags.
- Category Strength: Electronics outperform Apparel across both customer tiers.
- Strategic Leverage: Regular Electronics shows potential for premium migration.

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Qualitative Analysis

Qualitative Analysis

- Data analysis, especially relevant when working with interviews, open-ended surveys, group observations, or retail feedback
  - **Purpose**: Analyzes non-numeric data to understand concepts, opinions, or experiences. Context rich Interpretations that captures how and why of human responses.
  - **Examples**: Content analysis, thematic analysis, and narrative analysis.
  - **Use Case**: Analyzing open-ended survey responses, interviews, user experience studies and social media content.

Segment-wise Behavioral Insights

Premium Electronics

- Narrative: This segment reflects aspirational buying and brand loyalty. Customers here likely value innovation, exclusivity, and performance.
- Behavioral Signal: Consistent monthly growth suggests repeat purchases or successful product refresh cycles.
- Strategic Implication: Reinforce premium positioning with early-access launches, tech bundles, and loyalty tiers.

Premium Apparel

- Narrative: Customers may be style-conscious but selective. Growth is steady but not explosive.
- Behavioral Signal: Indicates seasonal or trend-driven buying, possibly influenced by fashion cycles or social proof.
- Strategic Implication: Introduce limited editions, influencer tie-ins, and curated collections to spark urgency.

Regular Electronics

- Narrative: This segment represents value-driven adoption. Customers are pragmatic and responsive to affordability and utility.
- Behavioral Signal: Accelerating growth suggests rising trust and product relevance.
- Strategic Implication: Promote reliability, offer EMI schemes, and highlight feature-value balance.

Regular Apparel

- Narrative: This segment may be saturated or undifferentiated. Customers buy out of necessity, not desire.
- Behavioral Signal: Slower growth and flattening trend suggest fatigue or lack of novelty.
- Strategic Implication: Reposition with fresh styles, bundle with electronics, or introduce loyalty discounts.

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Cognitive Analysis

Cognitive & Realtime Analytics

An advanced form of data analysis that mimics human cognition—such as learning, reasoning, and decision-making—while processing live data streams.

Blends AI, machine learning, and natural language processing to interpret complex, unstructured data instantly and contextually.

Features:

- Contextual Understanding:
  - Interprets meaning, intent, and relationships beyond raw numbers
- Adaptive Decision-Making:
  - Learns from feedback and adjusts recommendations or actions dynamically
- Multimodal Input Handling:
  - Works with text, images, audio, and video simultaneously

Examples:

- Fraud Detection:
  - Identifies suspicious behavior in financial transactions instantly
- Personalized Retail Offers:
  - Recommends products based on live browsing and purchase behavior

Example

Example: Battery Life Estimation

1. Descriptive Statistics

- Engineers test **1,000 smartphone batteries** and summarize their performance.
- They calculate **mean battery life, median, mode, and standard deviation** to describe overall trends.
- Result: **Average battery life = 10 hours, standard deviation = 1.5 hours.**

2. Inferential Statistics

- Instead of testing every battery ever made, engineers **analyze the sample data** and make predictions.
- Using **confidence intervals and hypothesis testing**, they **estimate the expected battery life for all future phones** based on this sample.
- Conclusion: "We are **95% confident** that the average battery life for all smartphones of this model will be **between 9.5 and 10.5 hours**."

Summary

- Discussed types of Analysis
- Emphasised the need and use of Descriptive Analytics
- Studies Predictive Analytics an its use
- Identified Other Analytics Techniques

Questions?